

MICROELECTROMECHANICAL SYSTEMS AND SENSORS

MEMS structures, transduction, fabrication and packaging

Lecture 4 of 16 · 80 minutes · Module A: Foundations

Two weeks ago a question had no answer. Today you learn why — and what to do instead.

Last week's muddiest points

From Lecture 3's exit tickets — answered before we start

1

INSTRUCTOR: fill these three in from the Lecture 3 exit tickets before class.

2

3

Refer to concepts, never to students. Ninety seconds total, then move on.

Retrieval, unaided

Lecture 1's scaling laws — say them before I show them

$$k \propto L$$

stiffness falls,
but only linearly

$$m \propto L^3$$

mass collapses —
the cube is the killer

$$f_0 \propto 1/L$$

resonance climbs:
this is why MEMS are fast

$$A/V \propto 1/L$$

the world becomes
all surface

Shrink a device and it gets faster and cheaper. It also gets lighter — and lighter means noisier.

Today those four relations stop being physics and become manufacturing. Every one of them is a consequence of how the device was made.

**“What is this device's
cross-axis sensitivity after it
has been soldered to my board?”**

Lecture 2, slide 22. The answer was: it is not in the datasheet, and it cannot be. I left that unpaid on purpose. Today it gets paid.

Why no manufacturer can answer it

Because the specification is not describing a component

1

A silicon structure a few micrometres thick

2

Suspended over a sealed cavity, free to move

3

Glued into a plastic box with an adhesive that cures and shrinks

4

Soldered to a board that flexes when you tighten a screw

The manufacturer owns line 1 and part of line 3. Lines 3 and 4 are yours — and they are where the error enters.

In Lecture 2 you were asked for a device's cross-axis sensitivity after it is soldered to your board, and the answer was that the number is not in the datasheet. Why is it not there?

- A** It is proprietary — competitors would learn from it
- B** It depends on your board and your assembly, so no manufacturer can specify it
- C** It is in the application note, not the datasheet
- D** It is the same as the package-level figure, so printing it twice would be redundant

Commit now. I am not telling you the answer until the end of the lecture — by which point you will have drawn the reason yourselves.

The packaging tax, in the datasheet you already own

ISM330DHCX · DS13012 Rev 6 · read the conditions column first

Parameter	Typ	Max	Unit	Conditions
Zero-g level	±10	±65	mg	25 °C, after soldering
Zero-g temp. coefficient	±0.1	±0.5	mg/°C	−40 to +85 °C
Cross-axis sensitivity	±0.5	—	%	25 °C, package level, not board level

“after soldering” — the manufacturer is telling you, in the specification itself, that your assembly is part of the error.

- ▶ Row 1 is an OFFSET: the output when the input is zero. Row 2 is DRIFT: how that offset moves.
- ▶ Row 3 is the number Lecture 2 asked for — and its condition line admits it is not yours.

Against Lecture 2's 8.73 mg

The number on the board since week two

$$\text{a } 0.5^\circ \text{ tilt} = \sin(0.5^\circ) \times 1 \text{ g} = \mathbf{8.73 \text{ mg}}$$

— the entire signal we are trying to measure

TYP ZERO-G LEVEL

$$\pm 10 \text{ mg} \div 8.73 \text{ mg}$$

1.15×

MAX ZERO-G LEVEL

$$\pm 65 \text{ mg} \div 8.73 \text{ mg}$$

7.45×

Even the TYPICAL offset exceeds the whole quantity we are trying to measure. The maximum is seven and a half times it.

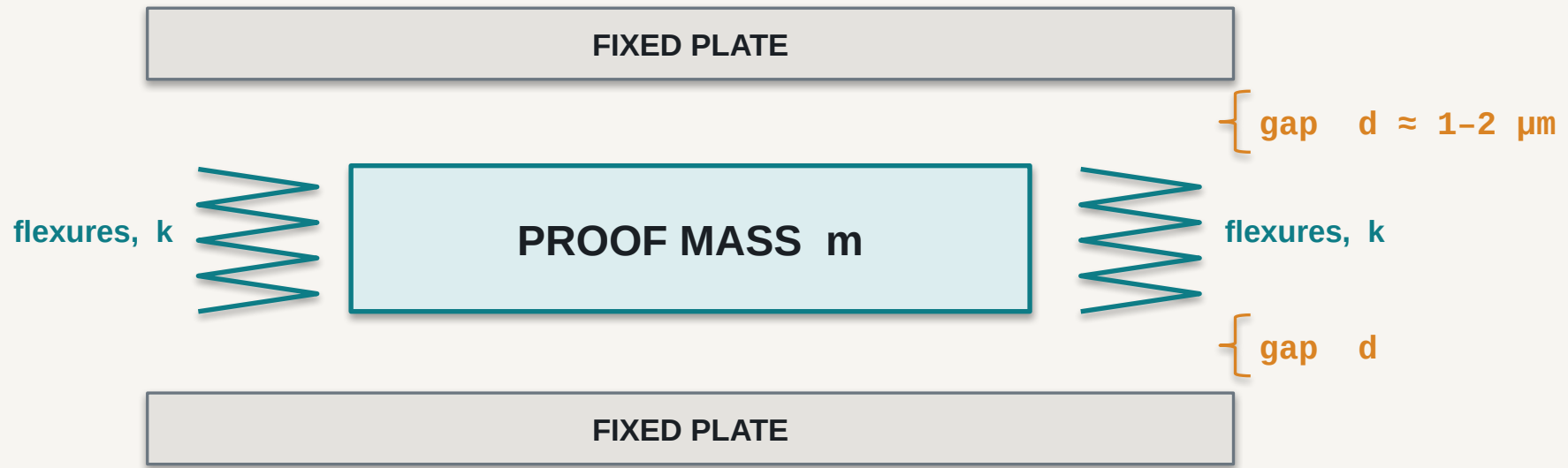
Five structures do all the work

Proof mass on a flexure. Cantilever beam. Diaphragm. Resonator. Comb.

Five shapes cover every device in Lectures 5 to 11. Learn the shapes, not the catalogue.

Structure 1, and you have seen it before

A capacitive MEMS accelerometer, in cross-section — Lecture 1's figure, unchanged



Acceleration moves the mass, one gap grows and the other shrinks, and the differential $C = \epsilon A/d$ is the signal.

In week one this was a drawing. Today it is a thing somebody etched, released, glued into a box and soldered down — and every one of those verbs changes **d**.

Proof mass on a flexure: the three relations

Everything an accelerometer datasheet says is a consequence of m , k and d

$$F = ma$$

the measurand becomes a force —
this is the only physics

$$x = ma / k$$

the force becomes a displacement —
the flexure sets the sensitivity

$$f_0 = (1/2\pi)\sqrt{(k/m)}$$

and the same m and k set
the usable bandwidth

One structure, one measurand, two specifications you cannot choose independently: sensitivity and bandwidth share m and k .

This is why the datasheet has selectable full-scale ranges but not selectable sensitivity at a fixed range: the mass and the springs were fixed by a mask set, once, for the whole production run. Range is switched in the electronics. The mechanics are not negotiable.

Structures 2 and 3

Cantilever beam · diaphragm over a cavity

CANTILEVER BEAM

$$k \propto E \cdot w \cdot t^3 / L^3$$

SENSES force, and it is the flexure above

IN THIS COURSE L8 — force and tactile sensors

Note the cube on t: thickness is set by DEPOSITION, so a 5 % film error is a 16 % stiffness error.

DIAPHRAGM OVER A CAVITY

$$\text{deflection} \propto \Delta P \cdot a^4 / (E \cdot t^3)$$

SENSES pressure, sound

IN THIS COURSE L8 pressure · L11 microphone

No proof mass anywhere in this device. Remember that at the next poll.

The beam and the flexure are the same object. Structure 2 is not a new idea — it is structure 1 with the mass taken off the end.

The diaphragm is the first structure that needs a SEALED CAVITY underneath it — which is why pressure sensors are packaging problems before they are electronics problems.

Structures 4 and 5

Resonator · comb — and the comb is the reason gyroscopes exist

RESONATOR

f_0 shifts with mass, stress, T

SENSES mass, gas, temperature, time

IN THIS COURSE L9 gas sensing · timing references

You do not measure an amplitude here. You measure a
FREQUENCY — which is the cheapest thing to measure well.

COMB STRUCTURE

$C = n \cdot \epsilon \cdot A/d$ $F \propto n \cdot V^2$

SENSES displacement — and it applies force too

IN THIS COURSE L5 · L6 · L10 MEMS mirror

The only structure that both SENSES and DRIVES.

A drive comb sustains a vibration. A sense comb reads the Coriolis deflection of that vibration. That is a gyroscope — and it is why one exists at all.

n is the number of finger pairs. The gap d is set by lithography — so the smallest feature a fab can print sets the capacitance, and that sets the noise floor.

The five structures, in one table

This is the whole of Chunk 1 — it is in the handout

Structure	Governing relation	Senses	In this course
Proof mass on a flexure	$x = ma/k, \quad f_0 = (1/2\pi)\sqrt{(k/m)}$	acceleration, angular rate	L5 accelerometer, L6 gyroscope
Cantilever beam	$k \propto E \cdot w \cdot t^3 / L^3$	force (it is the flexure above)	L8 force / tactile
Diaphragm over a cavity	deflection $\propto \Delta P \cdot a^4 / (E \cdot t^3)$	pressure, sound	L8 pressure, L11 microphone
Resonator	f_0 shifts with mass, stress, T	mass, gas, temperature, time	L9 gas microheater, timing
Comb structure	$C = n\epsilon A/d, \quad F \propto n \cdot V^2$	displacement — sense AND drive	L5, L6, L10 MEMS mirror

Five, not a taxonomy. Five is what fits in working memory — and five is genuinely enough for seven lectures.

Four MEMS devices. Which one does NOT contain a proof mass?

- A a 3-axis accelerometer
- B a vibrating-structure gyroscope
- C a barometric pressure sensor
- D a MEMS microphone

Vote alone, hands up on my count. Then I am going to tell you something honest about this question.

Four MEMS devices. Which one does NOT contain a proof mass?

A a 3-axis accelerometer

B a vibrating-structure gyroscope

C a barometric pressure sensor

D a MEMS microphone

If you voted D you were right about the physics and I asked a bad question — a microphone is a diaphragm too. Both C and D respond to pressure across a membrane, not to the inertia of a suspended block.

Why five shapes cover seven lectures

Every device in Module B, and the structure underneath it

Lecture	Device	Structure	Transduction
L5	accelerometer	proof mass on a flexure	capacitive
L6	gyroscope	comb drive + proof mass	capacitive
L7	magnetometer	none — nothing moves	electromagnetic
L8	pressure sensor	diaphragm over a cavity	piezoresistive
L8	force / tactile	cantilever beam	piezoresistive
L9	gas sensor	resonator / microheater	thermal
L10	MEMS mirror	comb	electrostatic drive
L11	microphone	diaphragm	capacitive

Eight devices. Five shapes. One of them needs no moving structure at all — and knowing which is worth as much as knowing the rest.

You are not about to learn eleven devices.

You are about to learn five structures and six transduction principles, and then watch them get recombined seven times. Module B is not a catalogue. It is one method, applied.

One structure, four different sensors

The resonator, to show that the shape is the reusable part

MASS	GAS	TEMPERATURE	TIME
a molecule lands on it; f_0 drops	a coating adsorbs a species; the mass changes	Young's modulus changes; f_0 tracks it	nothing changes; f_0 IS the output

Same beam. Same equation. Four products — and one of them is a bug in the other three.

Temperature shifts f_0 whether you asked it to or not: in a gas sensor that is an error term to compensate, and in a thermometer it is the whole product.

Six principles, one question that sorts them

Capacitive · piezoresistive · piezoelectric · thermal · electromagnetic · optical

The question: can it measure something that is not moving?

The two that carry Module B

Both answer yes to the static question

CAPACITIVE

STATIC (DC) RESPONSE: YES

SIGNAL

small, high-impedance

MAIN WEAKNESS

needs on-chip electronics; stray capacitance everywhere

EXPLAINS THE DATASHEET LINE

why the ISM330DHCX can measure tilt at all

PIEZORESISTIVE

STATIC (DC) RESPONSE: YES

SIGNAL

millivolts from a bridge

MAIN WEAKNESS

a strong temperature coefficient — the bridge drifts

EXPLAINS THE DATASHEET LINE

why pressure sensors need temperature compensation

Capacitive measures a GAP. Piezoresistive measures a STRAIN. Both hold their reading when the world stops moving — and that is not a given.

The one that answers no

Piezoelectric — self-generating, and blind to anything still

PIEZOELECTRIC

STATIC (DC) RESPONSE: NO

SIGNAL

charge, self-generating — no supply needed

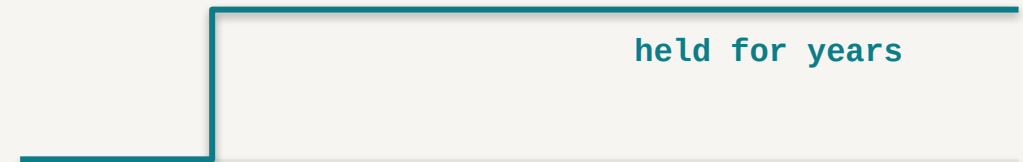
MAIN WEAKNESS

no DC response at all; the charge leaks away

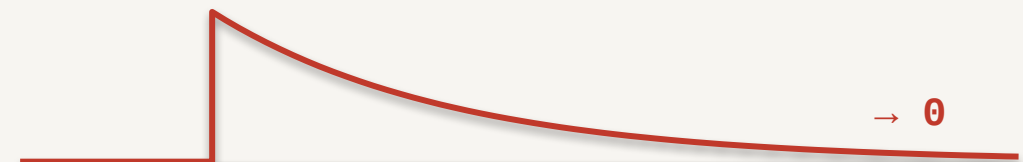
EXPLAINS THE DATASHEET LINE

why vibration sensors quote a LOW-frequency limit

TILT (the input) — stepped to 0.5° and held



PIEZOELECTRIC OUTPUT — decays with the amplifier's τ



A piezoelectric element responds to a CHANGE in strain. Hold it still and the output goes to zero — not badly, but exactly to zero.

Thermal and electromagnetic

Honestly brief — you meet both again in Module B

THERMAL

STATIC (DC) RESPONSE: YES, SLOWLY

SIGNAL

microvolts, or a resistance change

MAIN WEAKNESS

slow; and it heats itself, so it measures its own power

EXPLAINS THE DATASHEET LINE

the warm-up time on a gas sensor

ELECTROMAGNETIC

STATIC (DC) RESPONSE: YES

SIGNAL

large, low-impedance — the easy signal

MAIN WEAKNESS

hard to shrink, and it responds to every magnet nearby

EXPLAINS THE DATASHEET LINE

the hard- and soft-iron terms in Lecture 7

**Thermal is the only principle whose own operation disturbs the measurand.
Electromagnetic is the only one that fights the scaling laws instead of using them.**

Optical

The last principle, and the one that costs the most package

OPTICAL

STATIC (DC) RESPONSE: YES

SIGNAL

can be very large — sometimes no amplifier at all

MAIN WEAKNESS

needs a window, and the window sees ambient light

EXPLAINS THE DATASHEET LINE

the cover-window design in Lecture 10

THE THREE BRIEF ONES COME BACK HERE

L7

electromagnetic

magnetometers, and why they need calibrating in situ

L9

thermal

gas sensing, self-heating and warm-up

L10

optical

MEMS mirrors, and the window you must design

Three principles, one slide each. That is proportionate.

Optical is the only principle where the PACKAGE is part of the transduction: no window, no measurement.

A piezoelectric accelerometer — representative industrial vibration type — is mounted on the solar-tracker frame from Lecture 2 to measure its tilt. The frame is tilted to 0.5° and held there.

Thirty seconds later, the reported tilt is:

A 0.5° , correctly

B 0°

C 0.5° but very noisy

D it depends on the amplifier's gain

Vote. Then in pairs: one of you argues it reads 0.5° , the other argues it reads zero — then decide. Two minutes.

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D it depends on the amplifier's gain

Zero. Charge appears when the strain CHANGES; held still, it leaks away through the amplifier's input impedance in a few seconds. A piezoelectric accelerometer has no DC response. It cannot measure tilt — not badly, at all.

Six principles, sorted by one question

Can it measure something that is standing still?

Principle	Static (DC)?	Signal	Main weakness
Capacitive	yes	small, high-impedance	needs on-chip electronics; stray C
Piezoresistive	yes	mV from a bridge	strong temperature coefficient
Piezoelectric	no	charge, self-generating	no DC response; the charge leaks
Thermal	yes, slowly	μ V or resistance	slow; self-heating
Electromagnetic	yes	large, low-impedance	hard to shrink; magnetically susceptible
Optical	yes	can be very large	needs a window; ambient light

One “no” in the whole column — and it belongs to the principle that looks most impressive on a front page.

And what each one explains in a datasheet

The reason this chunk exists: every principle is a specification's cause

Principle	Static (DC)?	The datasheet line it explains
Capacitive	yes	why the ISM330DHCX can measure tilt at all
Piezoresistive	yes	why pressure sensors need temperature compensation
Piezoelectric	no	why vibration sensors quote a low-frequency limit
Thermal	yes, slowly	the warm-up time on a gas sensor
Electromagnetic	yes	the hard- and soft-iron terms in Lecture 7
Optical	yes	the cover-window design in Lecture 10

A specification is never arbitrary. Somewhere behind every line is a structure, a principle, or a process step.

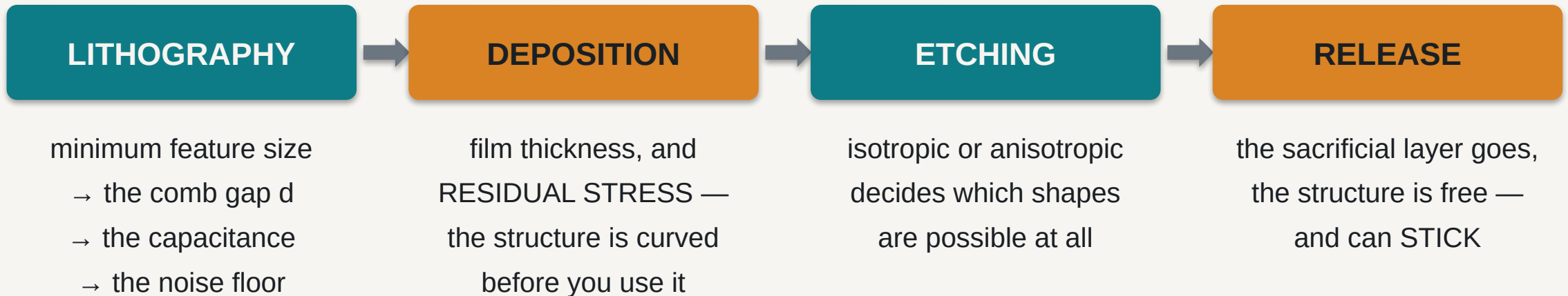
How it is made, and what that costs you

Four process steps. One consequence each. That is the whole treatment.

This is not a fabrication course — fabrication is here because it explains the datasheet.

Four steps, in order

One consequence each — and that is the entire treatment



PAIRS · 3 MINUTES · cards on your desk: put the four steps in order, then match each one to its consequence.

Then one question: which of the four is the reason a MEMS accelerometer is noisier than a bench instrument?

Bulk or surface

Two ways to build the same shape, and the trade-off is mass

BULK MICROMACHINING

- ▶ etches INTO the wafer
- ▶ thick structures, large proof masses
- ▶ low noise floor
- ▶ big die, expensive, hard to integrate
- ▶ pressure sensors, high-grade inertial

SURFACE MICROMACHINING

- ▶ builds UP from thin films
- ▶ micrometres thick, small masses
- ▶ higher noise floor
- ▶ cheap, small, on the same die as the electronics
- ▶ consumer IMUs — the part in your kit

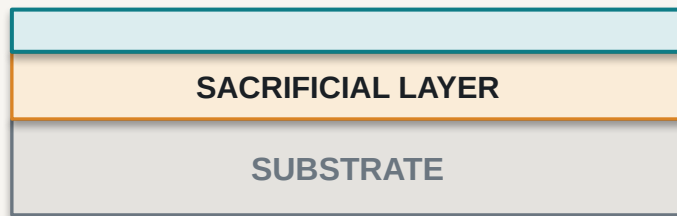
The trade-off is MASS — and by Lecture 1's $m \propto L^3$ you could already predict which one is quieter.

Your kit's part is not the quietest accelerometer buildable — only the quietest that fits beside its own ADC.

Release: the step that makes it a machine

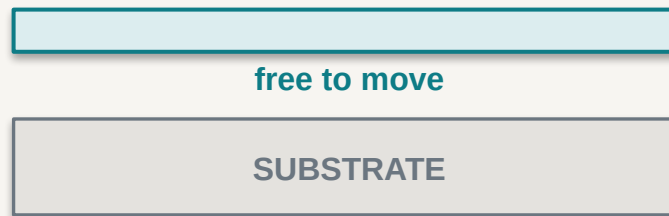
The sacrificial layer — and what happens when the beam comes down instead

1 · AS DEPOSITED



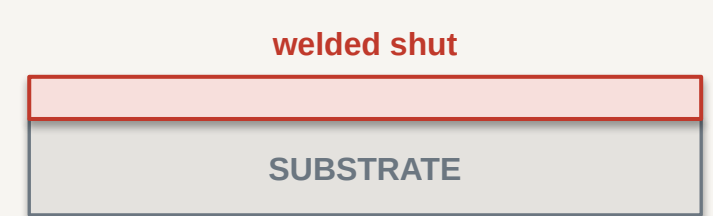
structural film on top of a sacrificial layer — nothing moves yet

2 · RELEASED



the sacrificial layer is etched away.
The beam is free. This is the machine.

3 · STUCK



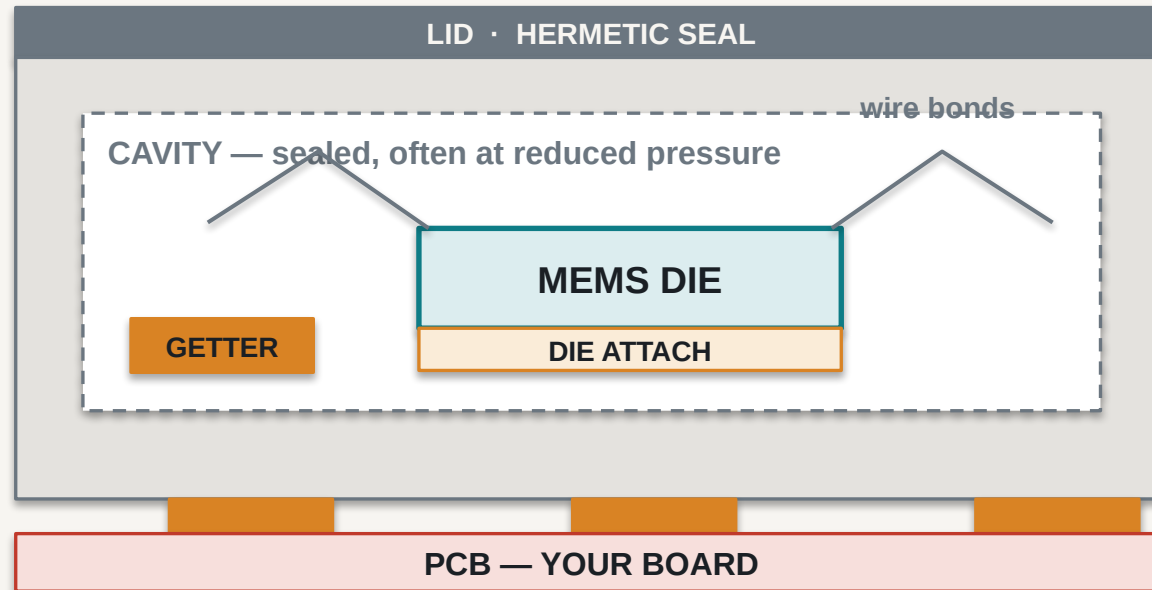
surface forces pull it down and it stays down. The device is dead.

Stiction is Lecture 1's $A/V \propto 1/L$ arriving as a yield problem: at this scale surface forces beat the restoring force of the spring.

Stiction is also a field failure: condensation, shock or electrostatics can bring a beam down later.

The package is not protection. It is physics.

Cavity · die attach · wire bond · seal · getter



WHAT EACH PART IS ACTUALLY FOR

CAVITY	a void at a controlled pressure — and that pressure sets the damping
DIE ATTACH	an adhesive that cures, shrinks and stresses the die on day one
WIRE BOND	a mechanical link to a moving object: a spring, and a stress
SEAL	keeps dust out of a 1–2 μm gap where one speck is fatal
GETTER	absorbs gas that leaks in, so the damping stays as designed

Two of the five put the die under stress before you ever power it up. That stress is the “after soldering” in the conditions column.

Nothing on this slide is optional, and nothing on it is specified for your board.

Two tens of a milli-g, and they are not the same ten

Name this collision out loud — it is a guaranteed exam misconception otherwise

	Lecture 2's 10 mg	Lecture 4's 10 mg
What it is	offset DRIFT over temperature	zero-g OFFSET after soldering
Where it comes from	the temperature coefficient of the structure	die attach, package and solder stress
The arithmetic	$\pm 0.5 \text{ mg/}^\circ\text{C} \times 20 \text{ }^\circ\text{C} = 10 \text{ mg}$	$\pm 10 \text{ mg typ, } \pm 65 \text{ mg max}$
Against the 8.73 mg signal	1.15×	1.15× typ, 7.45× max
Can you fix it?	NO — it moves after you calibrate	YES — one-point calibration on your board
Cost of the fix	—	one measurement, once, per unit

Fabrication and packaging hand you a LARGE but REMOVABLE error. The temperature coefficient hands you a SMALL-LOOKING but IRREMOVABLE one.

That both land on ten is a coincidence. The DIFFERENCE between them is the thesis of Module A.

You are specifying a pressure sensor for the capstone project. Four numbers matter. Which one must you measure on your own assembly, because no datasheet can give it to you?

- A** the noise density at your chosen bandwidth
- B** the sensitivity at 25 °C
- C** the zero-offset shift caused by your enclosure clamping the sensor
- D** the total error band over the operating temperature range

Transfer question. Different measurand, same distinction you have been building for the last two chunks.

You are specifying a pressure sensor for the capstone project. Four numbers matter. Which one must you measure on your own assembly, because no datasheet can give it to you?

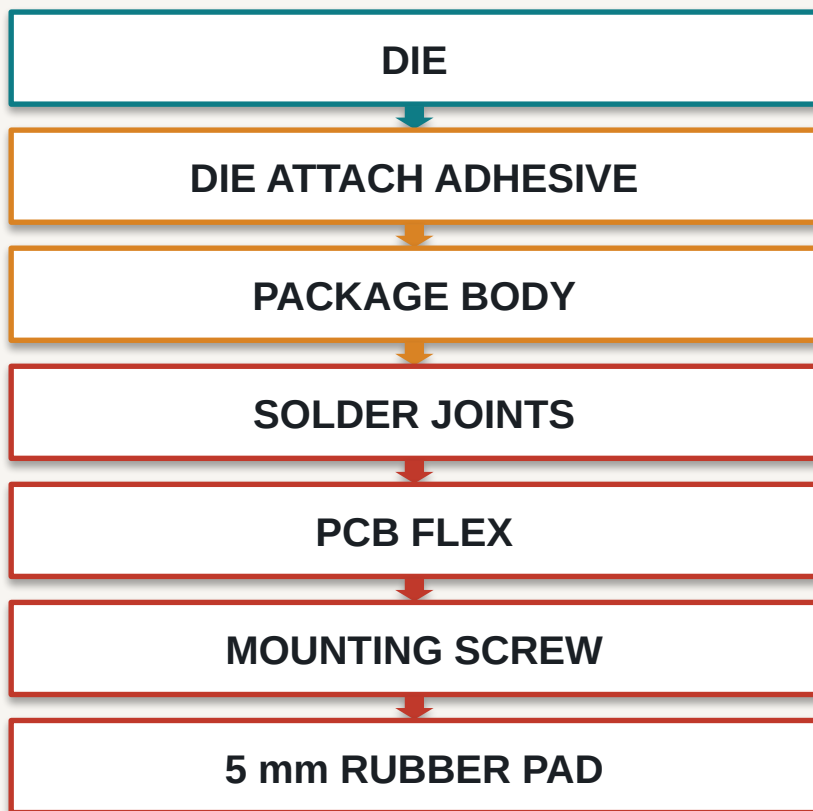
- A the noise density at your chosen bandwidth
- B the sensitivity at 25 °C
- C the zero-offset shift caused by your enclosure clamping the sensor**
- D the total error band over the operating temperature range

A, B and D are all in the datasheet — D is the very figure Lecture 2 taught you to prefer. C depends on your enclosure, your screw torque and your gasket, and it exists nowhere but on your own bench.

The stress chain

Seven links, and only the first is specified by anyone

THE CHAIN



WHAT IT CONTRIBUTES

the transduction itself

offset shift, hysteresis

the temperature coefficient

asymmetric stress → cross-axis error

offset and cross-axis, load-dependent

offset that changes when it is retightened

Lecture 1's war story, in one part

SPECIFIED BY ANYONE?

yes — the datasheet

partly — “after soldering”

partly

NO

NO

NO

NO

Lecture 1's story ended on the seventh link. Lecture 4 ends on the same picture.

In Lecture 2 you were asked for a device's cross-axis sensitivity after it is soldered to your board, and the answer was that the number is not in the datasheet. Why is it not there?

- A It is proprietary — competitors would learn from it
- B It depends on your board and your assembly, so no manufacturer can specify it**
- C It is in the application note, not the datasheet
- D It is the same as the package-level figure, so printing it twice would be redundant

Not secrecy, and not an application note. The chain you just drew has seven links and the manufacturer owns one of them. The number cannot exist — but the METHOD does: measure it on your own assembly.

The honest ledger of going small

Built with you — I ask for each row before I show it

Shrinking the device	buys you	costs you
$m \propto L^3$	—	less mass → a higher noise floor
$k \propto L$	—	—
$f_0 \propto 1/L$	higher bandwidth	resonance moves into your signal band
area / volume $\propto 1/L$	fast thermal response	surface forces dominate → stiction
batch fabrication	unit cost, integration with electronics	tolerances you cannot control
packaging	protection, handling	stress you cannot specify

The datasheet quantifies the middle column. The last two rows of the right-hand column are yours to measure.

Micro-scale is a trade, not an improvement.

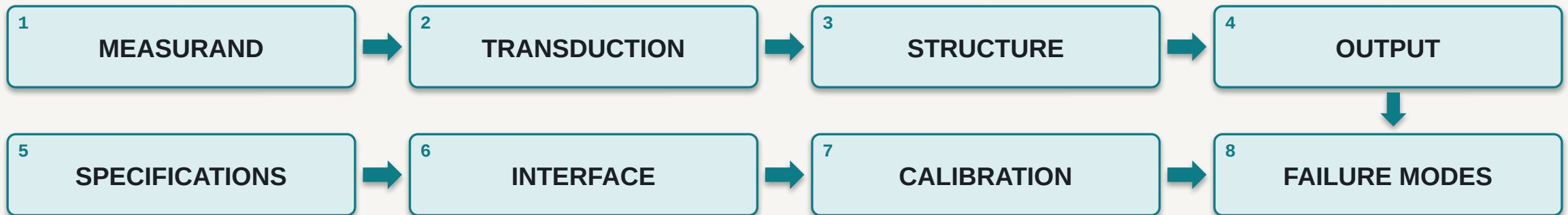
Everything you gained, you gained by giving something up. The datasheet quantifies the gains — and the last two rows of that ledger are yours to measure, on your own board, with your own screwdriver.

Module A ends here

Four lectures. One method. Now watch it get applied seven times.

You can now draw the chain, turn a request into a specification, read a datasheet against a requirement, and explain why a device behaves differently on your own board.

LECTURES 5–11 ARE SEVEN INSTANCES OF ONE PATTERN



Seven devices, one method. If you can state that pattern, the next seven weeks are one idea applied seven times rather than a list to memorise.

NEXT: Lecture 5 — the accelerometer in full: structure 1 and principle 1, down to a register value.

Exit ticket

Two questions · ninety seconds · handed in at the door

1 · YOUR PROJECT

Name one specification of the sensor in your capstone project that you will have to measure yourself — and say which stress path makes it necessary.

2 · THE PATTERN

Module A is over.

In one sentence: what is the pattern that Lectures 5–11 will repeat?

Question 2 is the one I am grading myself on. If fewer than half of you can state the pattern, Lecture 5 opens with it and I am not behind.

Reading: reader chapter 4 — structures and fabrication. Its failure section is the mounting-screw story.